

PROJECTION

FINAL PROJECT PITCH

INTERACTIVE MEDIA

GAME

Idea: Adaptive Projection Mapped Puzzle Game.

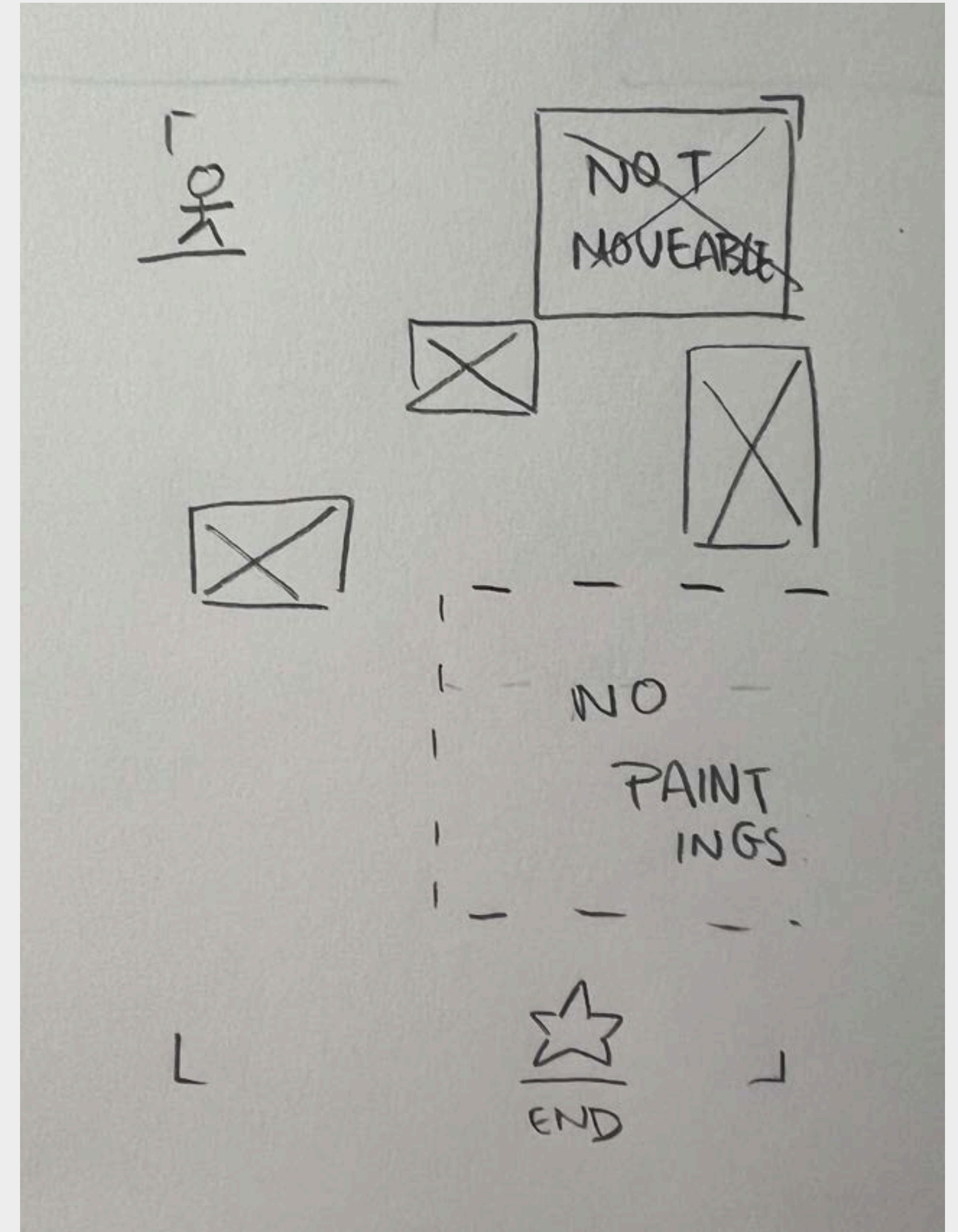
A puzzle game where the arrangement of picture frames determines the layout of the game terrain. When moved, the game layout will also change accordingly. The player's aim is to move their character to the goal by changing the arrangement of the picture frames in real time.

Different stages + Rules

There will be ~5 stages in the game.

The number of picture frames will remain the same each round however as levels increase:

- The number of moveable paintings will vary and some paintings will also be locked so they cannot be moved.
- Goal locations will vary.
- Blocked zones where paintings cannot be placed will be introduced in higher levels.
- Game area and size will also vary (some will be tighter than others)



Interactive irl game?

I love to play games.

Especially puzzle games that allow a bit more freedom but still require some creative input and strategising from players. (An example will be mentioned in my inspirations).

I am also unfortunately prone to migraines :(and can get easily triggered by the digital effects, like motion blur, in some games.

By taking away the digital screen and its implemented effects and utilising a projection mapping method (which i've always wanted to try), it feels as if I'm exploring a potential solution to my situation.

HOW: Tools

Projector

Project graphics and game on
to the picture frames and wall

Camera Sensor

Will sense when picture
frames have been moved
possibly through the detection of markers?

Learn where frames have
been relocated to.

Feed new
location to code

Code

Project/show coded
game through projector

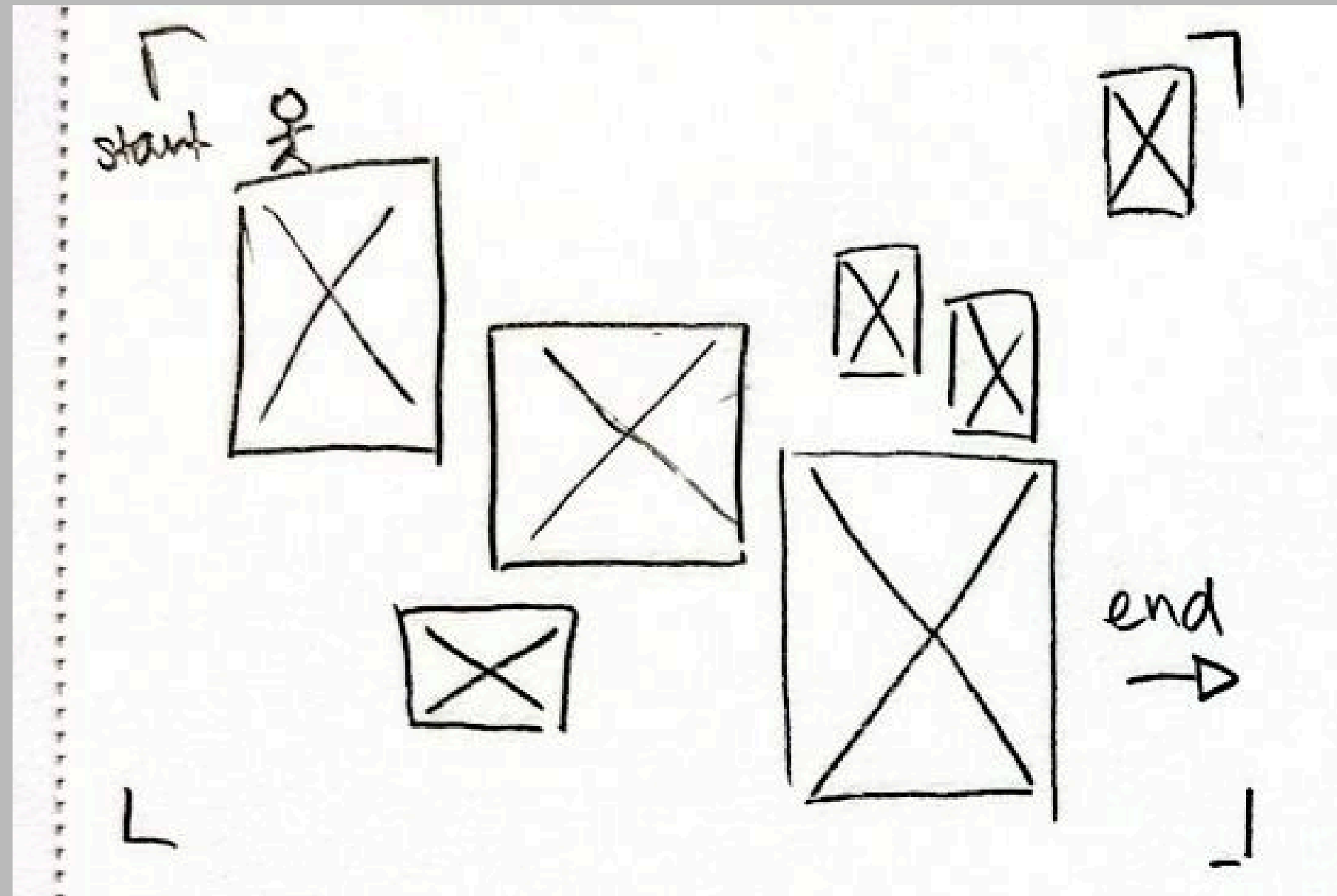
Learn where new picture
frames are located

Adjust boundaries
according to change and
project through projector

HOW

Code

the player must use keyboard/controller controls to navigate their way to the goal.

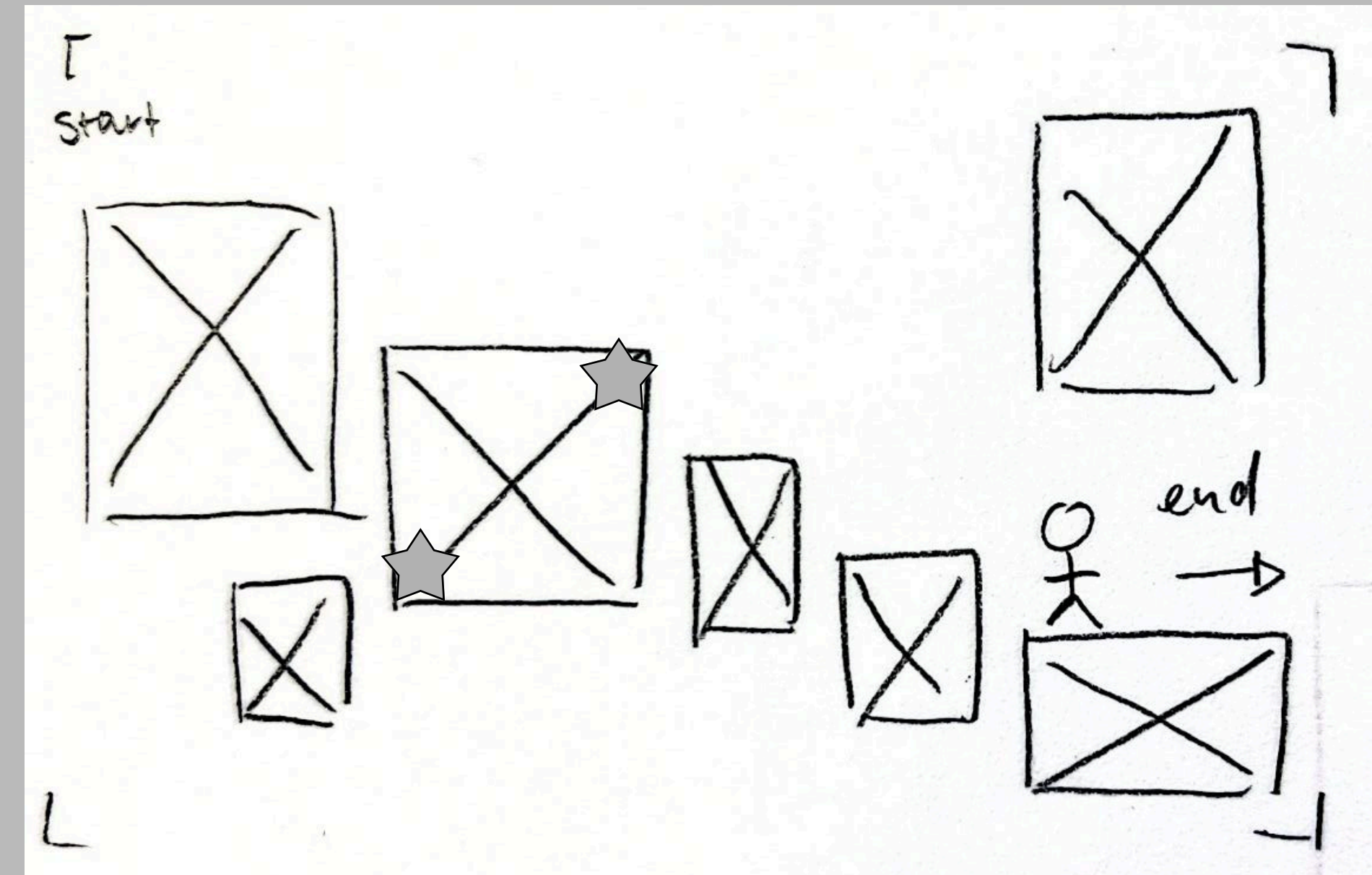


Projection Mapping

The game graphics will be projected onto the wall and picture frames so the picture frames become part of the terrain.

Camera Sensors

The placement of picture frames will be tracked through camera sensors to continuously update the game code.



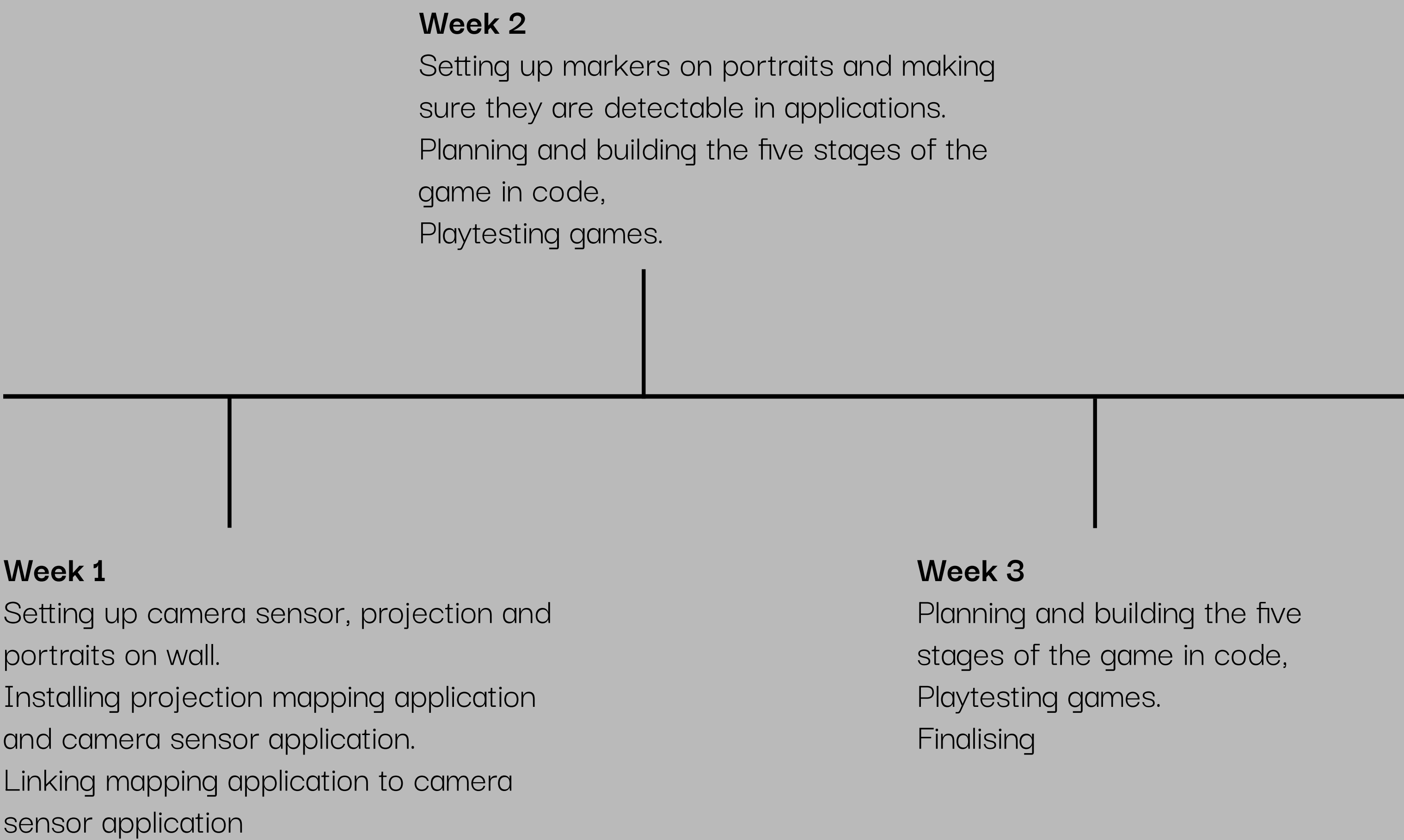
Human + Game interactivity

The interactive aspect will come from players physically rearranging the terrain to clear a path before using digital controls to move their character to the goal

HOW: Timeline

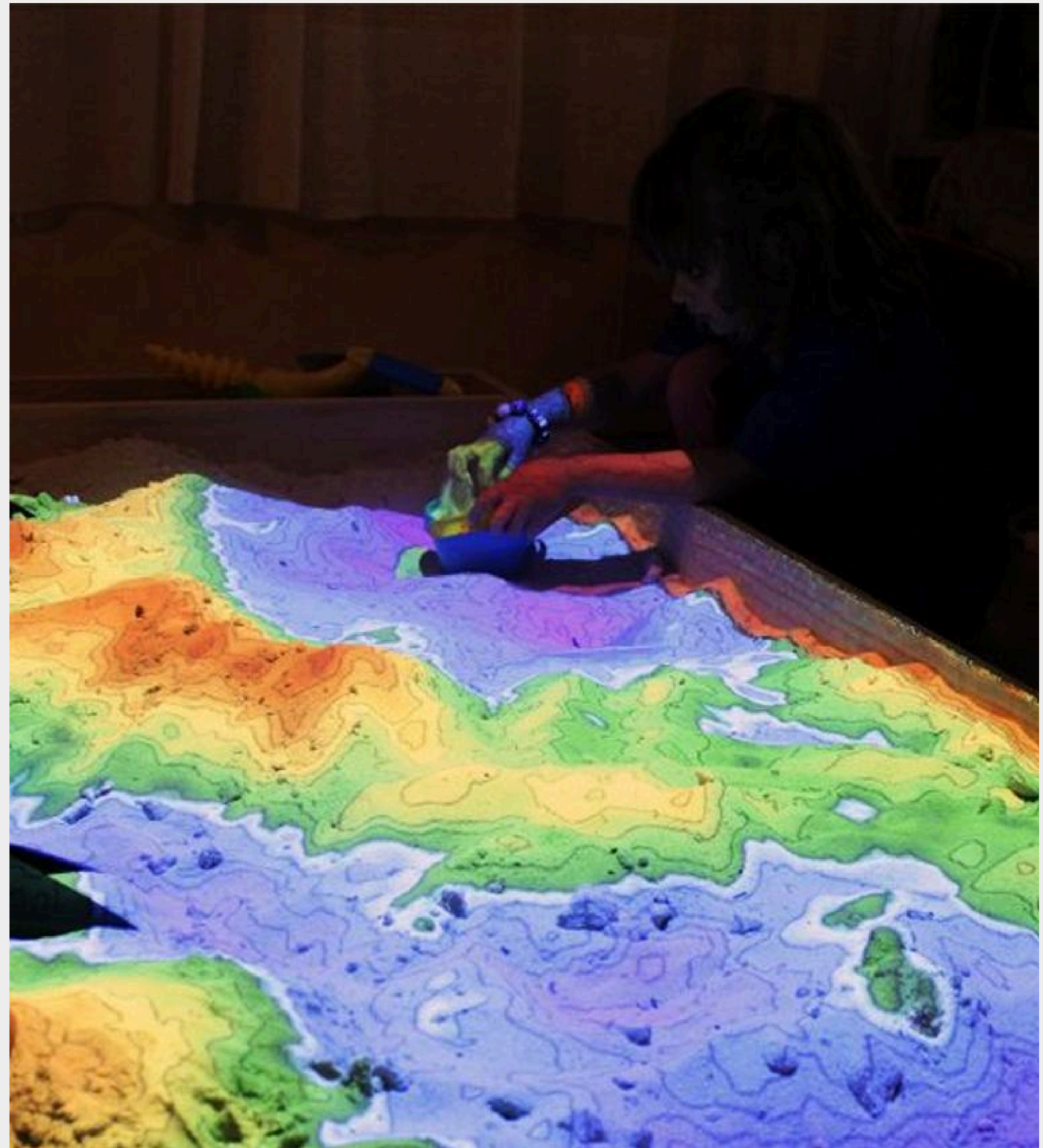
STEPS

- 1. Setting up camera sensor, projection and portraits on wall.
- 2. Installing projection mapping application and camera sensor application.
- 3. Linking mapping application to camera sensor application
- 4. Setting up markers on portraits and making sure they are detectable in applications.
- 5. Planning and building the five stages of the game in code,
- 6. Playtesting games.
- 7. Finalise and make sure everything is connected



Augmented Reality Sandbox

an interactive educational tool that combines a physical sandbox with real-time, 3D computer modeling. Using a Kinect camera and projector, it projects elevation colors and contour lines onto sand, enabling users to create, visualize, and interact with topography, water flow, and watersheds instantly.

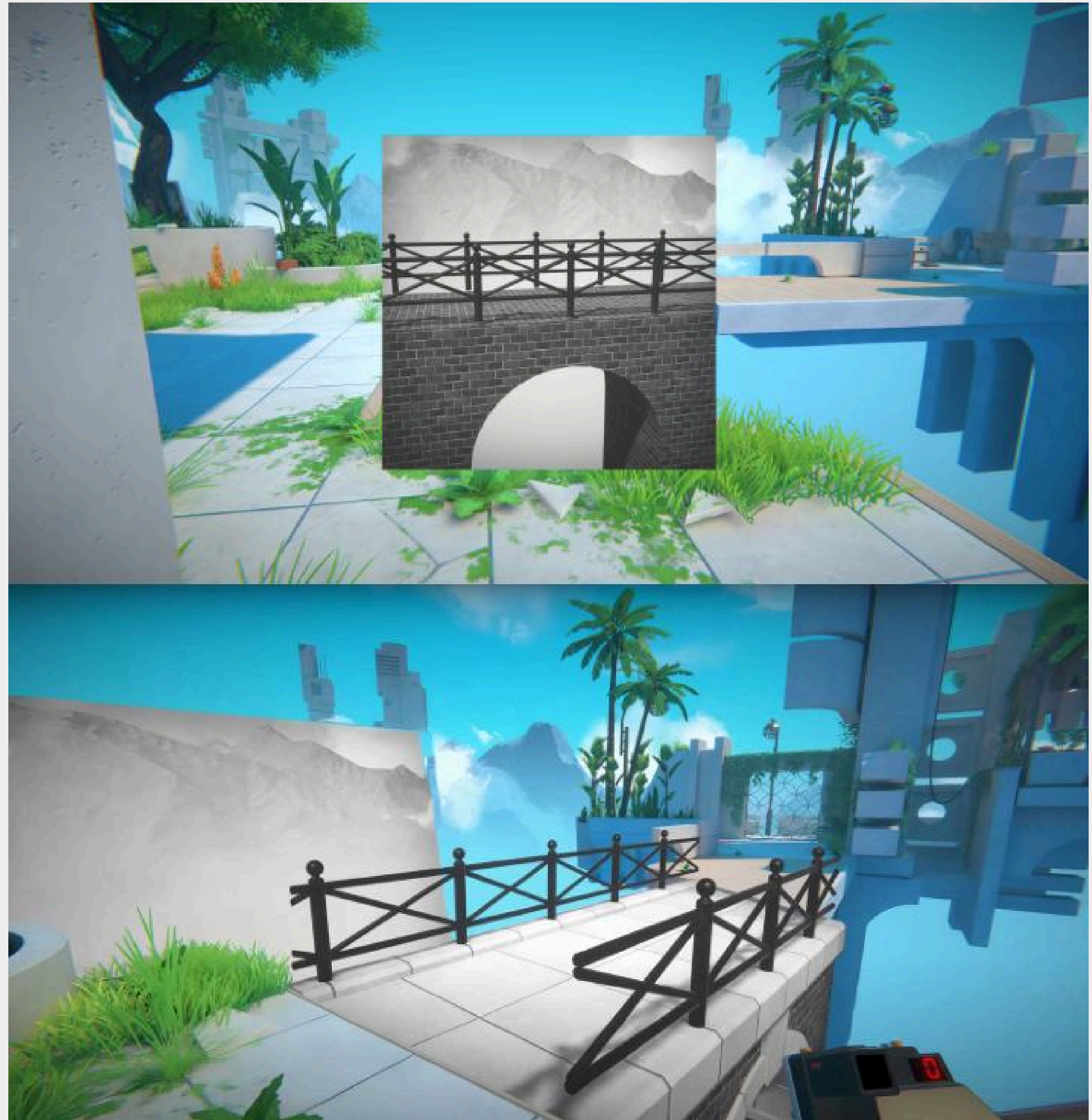


Game: Viewfinder

A game where you use an instant camera to reshape reality, placing photographs, sketches, and paintings into the world to make them 3D and traversable.

SAD OWL STUDIOS.

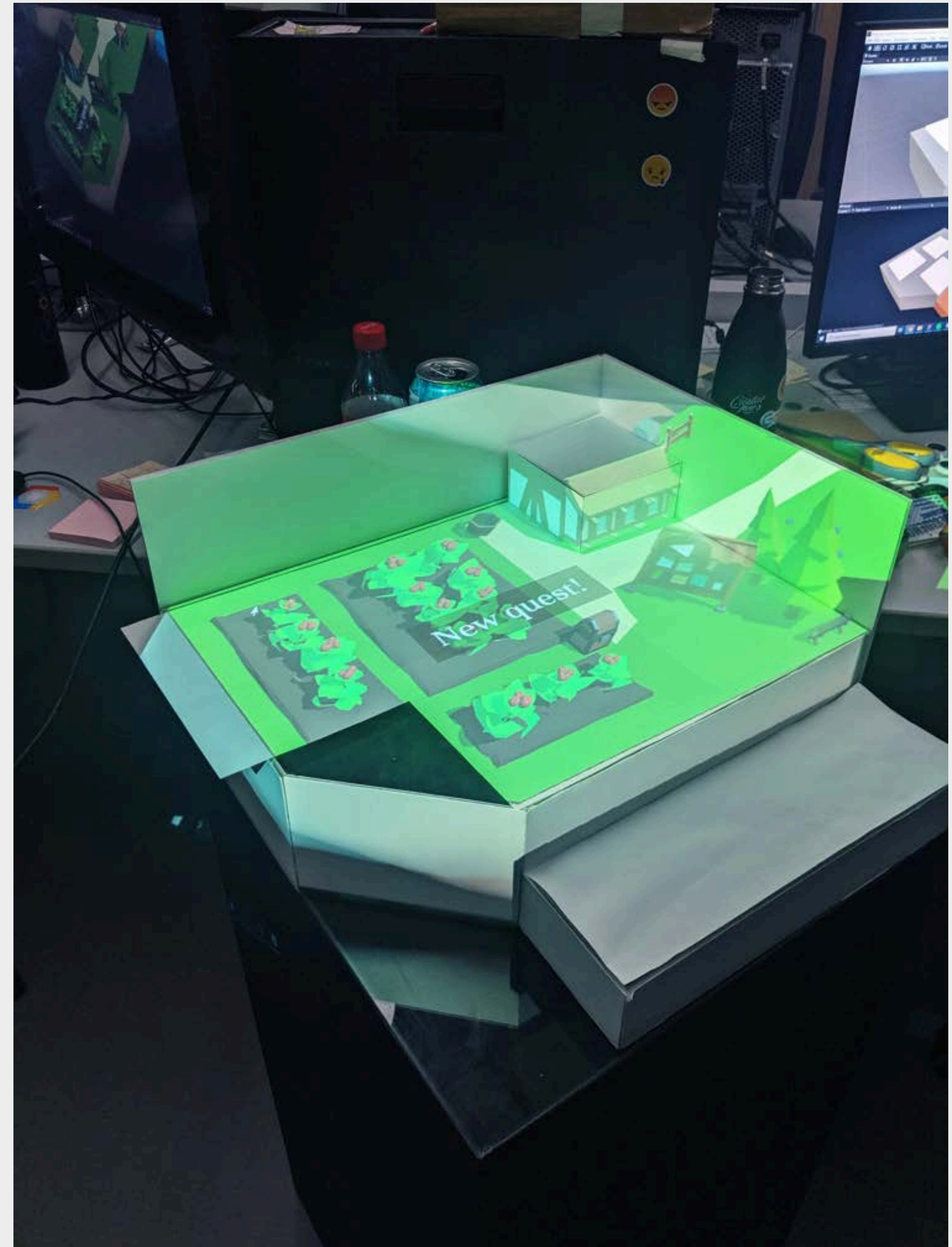
SPATIAL PUZZLES



Lightfarm

Users can touch the physical space to maintain a virtual farm and immediately visualize the impact of their decisions on soil quality, crop health, and water usage.

it uses projection mapping and computer vision to create natural, engaging interactions whilst teaching sustainable agricultural practices to smallholder farmers



Thank you